

Notes: Li, Wang, Cheung, and Jiang (RFS, 2011)

Evidence from the Split Share Structure Reform in China

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1 Big picture

This paper studies a large, economy-wide privatization episode in China—the *Split Share Structure Reform*—to answer a concrete question:

When a market friction is removed (here, a trading restriction on a large block of shares), what efficiency gain matters in equilibrium, and how is that gain reflected in the *negotiated* terms of privatization?

Before the reform, A-share firms had two types of domestic shares:

- **Tradable shares (TS):** held largely by domestic individuals and institutions, freely traded.
- **Non-tradable shares (NTS):** held by the state and legal-person entities, *not* freely traded.

The reform (2005–2007) converted NTS into tradable shares *gradually* after lockups. Crucially, conversion required a bargaining-and-voting process in which NTS holders compensated TS holders. The object of interest is therefore the **compensation ratio**:

$$\text{Compensation ratio} \equiv \frac{\text{additional shares received by TS holders}}{\text{each tradable share held}}.$$

The paper argues that compensation reflects *two* forces:

1. **Price impact:** bringing a large supply of previously illiquid shares to the market can depress prices, which TS holders care about.
2. **Risk-sharing gains:** NTS holders were often under-diversified ex ante (concentrated in one firm); tradability allows them to sell and diversify, so they are willing to pay to obtain this option.

The central empirical claim is that *risk-sharing gains* are a quantitatively meaningful determinant of compensation, above and beyond pure price-impact considerations.

2 Institutional setting and the bargaining timeline (Figure 1)

The reform proceeds as a structured bargaining game with an explicit veto by TS holders. A stylized timeline is:

1. t_0 : firm announces reform start; trading is suspended; the board (representing NTS holders) proposes an initial compensation plan.
2. Negotiation: TS and NTS holders communicate (internet voting, meetings, investor feedback). The plan can be revised (often upward).
3. t_1 : final plan is announced; trading resumes.
4. t_2 : registration date for voting eligibility; trading is suspended again shortly after.

5. Voting: TS holders vote. The plan must be approved by at least *two-thirds* of voting TS holders.
6. t_3 : reform implementation; compensation shares are transferred; trading resumes and compensation shares become tradable immediately for TS holders (while NTS shares face lockups).

This institutional structure matters because it makes compensation an *equilibrium outcome* of bargaining subject to (i) TS-holder approval, (ii) NTS-holder incentives to complete the reform, and (iii) market expectations that evolve across reform batches.

3 Economic mechanism: why risk sharing enters

A stock's total risk decomposes into systematic and idiosyncratic components. The paper's key observation is *asymmetric diversification* pre-reform:

- TS holders can trade and (in equilibrium) are more likely to hold diversified portfolios.
- Many NTS holders are effectively concentrated owners: they hold large stakes in one firm and cannot sell, so idiosyncratic risk is *personally relevant*.

Making NTS tradable gives NTS holders a diversification technology (they can sell to the market), thereby sharing idiosyncratic risk with the broad investor base. If that idiosyncratic risk is larger, the benefit from being able to diversify is larger, hence NTS holders are willing to pay more.

4 Hypotheses

The paper organizes predictions into five hypotheses:

- **H1 (Risk-sharing channel):** compensation is increasing in inferred gains from risk sharing.
- **H2 (Heterogeneity by NTS-holder diversification):** the sensitivity of compensation to risk-sharing gains is stronger when NTS holders are less diversified ex ante.
- **H3a (Price impact, quantity side):** compensation is increasing in the fraction of NTS (more shares potentially coming to market).
- **H3b (Price impact, valuation wedge):** compensation is increasing in S/PN , where S is the pre-reform tradable-share price and PN is the pre-reform value of an NTS share.
- **H4 (Bargaining power):** compensation is decreasing in NTS holders' bargaining power.
- **H5 (Firm performance):** compensation is decreasing in firm operating performance.

The paper's empirical design tests these jointly, and then compares them to a *pure price impact* benchmark model.

5 A pure price-impact benchmark model (Equations (1)–(3))

The benchmark deliberately *ignores* risk-sharing benefits and assumes the compensation ratio is set so that the *shareholding values* of TS and NTS holders are preserved across the reform.

5.1 Notation

Let

- $\pi \in (0, 1)$ be the fraction of shares that are tradable before the reform.
- S be the price of a tradable share before the reform.
- S' be the price of that share after the reform.
- PN be the value of a non-tradable share before the reform (e.g., proxied by net asset value per share).
- γ be the *compensation ratio* in shares: TS holders receive γ additional shares for each tradable share held.

5.2 Value-preservation conditions

TS holders. Before reform, TS holders own π shares per share outstanding at value S , so their per-share-of-firm value is πS . After reform, they still own their original π tradable shares *plus* receive $\gamma\pi$ additional shares, all valued at S' . Thus the model imposes:

$$\pi S = (\pi + \gamma\pi)S'. \quad (1)$$

NTS holders. Before reform, NTS holders own $(1-\pi)$ shares valued at PN , so value is $(1-\pi)PN$. After reform, they transfer $\gamma\pi$ shares to TS holders, leaving $(1-\pi-\gamma\pi)$ shares valued at S' , so:

$$(1-\pi)PN = (1-\pi-\gamma\pi)S'. \quad (2)$$

5.3 Solving for the implied compensation ratio

From (1), $S' = \frac{S}{1+\gamma}$. Plug into (2):

$$(1-\pi)PN = (1-\pi-\gamma\pi)\frac{S}{1+\gamma}.$$

Multiply both sides by $(1 + \gamma)$ and rearrange to isolate γ . A convenient way is to write everything in terms of S/PN :

$$\begin{aligned}
(1 - \pi)PN(1 + \gamma) &= (1 - \pi - \gamma\pi)S, \\
(1 - \pi)PN + (1 - \pi)PN\gamma &= (1 - \pi)S - \gamma\pi S, \\
\gamma((1 - \pi)PN + \pi S) &= (1 - \pi)(S - PN), \\
\gamma &= \frac{(1 - \pi)(S/PN - 1)}{(1 - \pi) + \pi(S/PN)}. \tag{3}
\end{aligned}$$

5.4 Comparative statics and testable implications

Assuming $S > PN$ (tradable shares are priced above the value assigned to non-tradable shares), the model predicts:

- higher $(1 - \pi)$ (a larger non-tradable block, hence bigger potential supply shock) $\Rightarrow$ higher γ (**H3a**),
- higher S/PN (a larger wedge between market price and NTS valuation) $\Rightarrow$ higher γ (**H3b**).

Importantly, this benchmark generates *no role* for risk sharing. The paper uses it as a falsifiable alternative.

6 Data, sample construction, and variable measurement

6.1 Sample

The authors start with 1,254 listed firms that completed reform by December 31, 2007, and keep firms with full data for the most comprehensive specification, yielding 1,147 firms (about 85% of A-share market capitalization at end of 2007). Continuous variables are winsorized at the 1st and 99th percentiles.

6.2 Key variables

Dependent variable: compensation ratio. Collected from final compensation plans (WIND database). Mean is about 0.301 shares per tradable share.

Risk-sharing proxy: idiosyncratic risk. Measured as the standard deviation of residuals from a market-model regression:

$$r_{i,t} = \alpha_i + \beta_i r_{m,t} + \varepsilon_{i,t}, \quad \text{IdioRisk}_i = \sqrt{\text{Var}(\varepsilon_{i,t})}.$$

Estimation window uses daily returns from 260 to 60 trading days prior to the first reform announcement; for newer firms, from one trading day after IPO to 60 trading days prior to announcement.

Price-impact proxy: fraction of NTS.

$$\text{Fraction of NTS}_i = \frac{\text{non-tradable shares}}{\text{total shares outstanding (pre-reform)}}.$$

Higher values mechanically imply a larger potential increase in float.

Bargaining power proxy: state NTS holding. State ownership (and state-controlled entities) among NTS holders, measured as a fraction of total shares. The maintained interpretation is that higher state ownership implies *weaker* bargaining power of NTS holders because the state has a strong incentive to complete the reform (political economy and social stability considerations).

Firm performance proxy: ROA. Return on assets, expected to reduce compensation because strong firms make NTS holders more willing to continue holding rather than sell.

Process controls.

- **Batch:** sequencing of firms in the reform process (learning and expectation effects).
- **Plan change:** indicator that the proposed plan was revised during negotiation.
- **Additional promise:** indicator that NTS holders offered additional commitments (lockups, dividend promises, performance guarantees, etc.).

Additional controls. The regressions add standard firm characteristics (sales growth, market-to-book, prior stock return, leverage), ownership structure (institutional TS holdings, foreign listing/issuance indicator), and governance (CEO also chair, board independence, board size).

7 Baseline empirical specification (Equation (4))

The main regression tests H1, H3a, H4, and H5 jointly:

$$\begin{aligned} \text{CompensationRatio}_i = & \alpha_0 + \beta_1 \text{IdioRisk}_i + \beta_2 \text{FractionNTS}_i + \beta_3 \text{StateNTSHolding}_i \\ & + \beta_4 \text{ROA}_i + \beta_5 \text{Batch}_i + \beta_6 \text{PlanChange}_i + \beta_7 \text{AdditionalPromise}_i \\ & + f_{\text{NTS holder}} + f_{\text{year}} + f_{\text{industry}} + \varepsilon_i. \end{aligned} \quad (4)$$

The inclusion of NTS-holder fixed effects is important: it differences out time-invariant heterogeneity across controlling NTS holders (e.g., sophistication, ownership style, political objectives).

8 Main results

8.1 Figure 1: what to take away

Figure 1 is not a statistical figure; it is the institutional map of the bargaining/voting process. The key takeaway for identification is that *TS holders have an explicit voting threshold*, so the

compensation ratio is a negotiated object that must clear a supermajority among TS holders.

8.2 Table 1: summary statistics and correlations

Table 1 shows substantial cross-sectional dispersion in compensation. Two correlations are most informative for the paper’s mechanism:

- Compensation is positively correlated with idiosyncratic risk (consistent with risk-sharing motives).
- Compensation is strongly positively correlated with the fraction of NTS (consistent with price-impact motives).

Table 1 also motivates the multivariate approach: several covariates (e.g., batch, state ownership, ROA) co-move with compensation.

Panel A: A General Case (single firm)				
t_0	t_1	t_2	t_3	
Announcement of the start of the reform	Negotiation results released	Registration date for voting	Announcement of the completion of the reform	
Trading suspended and negotiation takes place	Trading resumes	Trading suspended and voting takes place	Trading resumes	

Panel B: A Case Study – Shanghai Auto’s Split Share Structure Reform				
09-12-2005	09-21-2005	09-29-2005	10-24-2005	
Announcement of the start of the reform	Negotiation results released	Registration date for voting	Announcement of the completion of the reform	
Trading suspended and negotiation takes place	Trading resumes	Trading suspended and voting takes place	Trading resumes	

Figure 1
Time line: split share structure reform

Table 1 Summary statistics							
Panel A: Summary statistics							
Variable	N	Mean	Std Dev	25 th Percentile	Median	75 th Percentile	
Compensation ratio	1147	0.301	0.075	0.270	0.306	0.342	
Idiosyncratic risk	1147	0.021	0.006	0.017	0.020	0.024	
Fraction of NTS	1147	0.602	0.113	0.543	0.620	0.686	
State NTS holding	1147	0.340	0.265	0.021	0.379	0.587	
ROA	1147	0.048	0.086	0.021	0.046	0.086	
Batch	1147	23.956	15.737	13.000	22.000	33.000	
Plan change	1147	0.839	0.368				
Additional premise	1147	0.699	0.459				
Panel B: Correlation matrix							
	1	2	3	4	5	6	7
1 Compensation ratio	1.000						
2 Idiosyncratic risk	0.080***	1.000					
3 Fraction of NTS	0.433***	0.072**	1.000				
4 State NTS holding	0.271***	-0.131***	0.288***	1.000			
5 ROA	-0.014	-0.153***	0.130***	0.097***	1.000		
6 Batch	-0.260***	0.147***	-0.154***	-0.021	-0.401***	1.000	
7 Plan change	0.120***	-0.065**	0.031	0.209***	-0.030	0.092***	1.000
8 Additional premise	-0.087***	-0.191***	-0.033	0.021	0.164**	-0.181***	0.028

Table 2 The compensation ratio				
	(1)	(2)	(3)	(4)
Idiosyncratic risk	0.902*** [0.004]	1.004*** [0.006]	0.821** [0.013]	0.820** [0.013]
Fraction of NTS	0.220*** [0.000]	0.234*** [0.000]	0.220*** [0.000]	0.220*** [0.000]
State NTS holding	0.046*** [0.000]	0.045*** [0.000]	0.044*** [0.000]	0.045*** [0.000]
ROA	-0.182*** [0.000]	-0.142*** [0.001]	-0.086** [0.044]	-0.085** [0.046]
Batch	-0.001*** [0.000]	-0.001*** [0.000]	-0.002*** [0.000]	-0.001*** [0.000]
Plan change	0.024*** [0.001]	0.022*** [0.001]	0.021*** [0.001]	0.022*** [0.001]
Additional premise	-0.013*** [0.000]	-0.012*** [0.000]	-0.009** [0.034]	-0.009** [0.037]
Sales growth	-0.010* [0.080]	-0.008 [0.151]	-0.009 [0.139]	-0.009 [0.139]
M/B	-0.001 [0.786]	-0.000 [0.930]	-0.000 [0.880]	-0.000 [0.880]
Stock return	-0.016 [0.104]	0.004 [0.724]	0.004 [0.721]	0.004 [0.721]
Leverage	0.018 [0.245]	0.017 [0.265]	0.017 [0.265]	0.017 [0.278]
Institutional TS holding		-0.104*** [0.000]	-0.103*** [0.000]	-0.103*** [0.000]
Foreign investor		0.034*** [0.000]	0.034*** [0.000]	0.034*** [0.000]
CEO is COB		0.007 [0.293]	0.007 [0.293]	0.007 [0.293]
Board independence		0.046 [0.351]	0.046 [0.351]	0.046 [0.351]
Board size		0.002 [0.952]	0.002 [0.952]	0.002 [0.952]
Intercept	0.156*** [0.000]	0.143*** [0.000]	0.146*** [0.000]	0.130*** [0.000]
NTS holder, year, and industry FE	YES	YES	YES	YES
Number of observations	1147	1147	1147	1147
Adjusted R-squared	0.313	0.317	0.359	0.359

8.3 Table 2: baseline regression evidence for H1–H5

Table 2 estimates Eq. (4) with progressively richer controls.

H1 (risk sharing): $\beta_1 > 0$. Idiosyncratic risk enters positively and significantly across specifications (e.g., about 0.82–1.00 in the main columns), consistent with higher compensation when risk-sharing gains are larger.

H3a (price impact): $\beta_2 > 0$. The fraction of NTS has a large, precisely estimated positive coefficient (about 0.23–0.25). This is consistent with compensation being higher when more shares will eventually be supplied to the market.

H4 (bargaining power): $\beta_3 > 0$ with the paper’s sign convention. State NTS holding is positive and significant (about 0.044–0.046). Under the maintained interpretation (state wants completion), this means weaker NTS-holder bargaining power $\Rightarrow$ higher compensation.

H5 (performance): $\beta_4 < 0$. ROA is negative and (mostly) statistically significant. Interpreted structurally, better firms make NTS holders less willing to pay for immediate diversification and less willing to concede in bargaining.

Process variables. Batch has a negative coefficient (later batches have lower compensation, consistent with learning/expectations). Plan changes are associated with higher final compensation (revisions tend to move upward). Additional promises substitute for shares (negative coefficient).

Economic magnitudes (reported by the authors). Moving idiosyncratic risk from the 25th to the 75th percentile increases compensation by about 0.007 shares, while moving fraction NTS from the 25th to the 75th percentile increases compensation by about 0.033 shares. Thus, in this accounting, price impact is the dominant factor, but risk sharing contributes meaningfully.

8.4 Table 3: heterogeneity by NTS-holder diversification (H2)

The paper operationalizes diversification using the controlling NTS holder's breadth of holdings:

- **Diversified NTS holders:** controlling shareholders that hold shares in at least five listed firms.
- **Non-diversified NTS holders:** controlling shareholders below that threshold.

Table 3
Heterogeneity among NTS holders

Panel A: List of diversified NTS holders

Controlling NTS holders	Number of firms with NTS stock holdings	Total market capitalization (billion Yuan)	Average market capitalization (billion Yuan)	Fraction of their controlled share value over the total market capitalization
SASAC	143	1055.77	7.033	0.262
SASAC of Shanghai Municipal Government	43	167.807	3.884	0.043
SASAC of Beijing Municipal Government	14	44.015	3.144	0.011
SASAC of Shandong Province	12	75.481	6.290	0.020
SASAC of Anhui Province	11	43.017	3.911	0.011
SASAC of Shenzhen Municipal Government	11	30.303	2.755	0.008
SASAC of Hunan Province	10	22.738	2.274	0.006
SASAC of Shanxi Province	9	39.478	4.386	0.010
SASAC of Fujian Province	8	24.192	3.024	0.006
SASAC of Jiangsu Province	8	24.091	3.011	0.006
SASAC of Yunnan Province	8	15.636	1.954	0.004
SASAC of Qingdao Municipal Government	8	10.448	1.306	0.003
SASAC of Hubei Province	7	31.945	4.564	0.008
SASAC of Guangzhou Municipal Government	6	22.497	3.750	0.006
SASAC of Nanjing Municipal Government	6	8.171	1.362	0.002
SASAC of Dalian Municipal Government	5	19.759	3.952	0.005
SASAC of Gansu Province	5	7.598	1.520	0.002
SASAC of Guizhou Province	5	9.467	1.893	0.002
SASAC of Jiangsu Province	5	6.147	1.229	0.002
SASAC of Shaoyang Municipal Government	5	8.605	1.721	0.002
SASAC of Wuhan Municipal Government	5	7.15	1.430	0.002
SASAC of Xinjiang Province	5	8.23	1.646	0.002
SASAC of Chongqing Municipal Government	5	6.412	1.282	0.002
Total	344	1638.159	4.762	0.425

(continued)

Table 3
Continued

Panel B: Subsample of firms controlled by diversified NTS holders

	(1)	(2)	(3)	(4)
Idiosyncratic risk	0.303 [0.680]	0.310 [0.691]	0.638 [0.332]	0.638 [0.331]
Fraction of NTS	0.260*** [0.000]	0.265*** [0.000]	0.237*** [0.000]	0.237*** [0.000]
State NTS holding	0.066*** [0.000]	0.067*** [0.000]	0.054*** [0.017]	0.054*** [0.017]
ROA	-0.224*** [0.010]	-0.139* [0.065]	-0.096 [0.214]	-0.096 [0.218]
Batch	-0.001** [0.027]	-0.001 [0.027]	-0.001** [0.021]	-0.001** [0.021]
Plan change	0.013 [0.251]	0.012 [0.256]	0.007 [0.479]	0.007 [0.514]
Additional promise	-0.015** [0.025]	-0.012 [0.026]	-0.009 [0.085]	-0.009 [0.085]
Sales growth	-0.224*** [0.000]	-0.001 [0.900]	-0.096 [0.869]	-0.096 [0.899]
M/B	-0.004 [0.311]	-0.005 [0.219]	-0.005 [0.353]	-0.005 [0.353]
Stock return	-0.028* [0.090]	0.001 [0.961]	0.000 [0.985]	0.000 [0.985]
Leverage	0.039* [0.085]	0.037* [0.111]	0.036 [0.111]	0.036 [0.111]
Institutional TS holding	0.000 [0.000]	0.000 [0.000]	-0.009*** [0.000]	-0.009*** [0.000]
Foreign investor	0.025*** [0.000]	0.024*** [0.000]	0.024*** [0.000]	0.024*** [0.000]
CEO is COB	-0.006 [0.454]	-0.006 [0.454]	-0.006 [0.454]	-0.006 [0.454]
Board independence	0.023 [0.781]	0.023 [0.781]	0.023 [0.781]	0.023 [0.781]
Board size	0.000 [0.802]	0.000 [0.802]	0.000 [0.802]	0.000 [0.802]
Intercept	0.197*** [0.000]	0.173*** [0.000]	0.174*** [0.000]	0.165*** [0.000]
NTS holder, year, and industry FE	YES	YES	YES	YES
Number of observations	344	344	344	344
Adjusted R-squared	0.297	0.318	0.373	0.368

(continued)

Table 3
Continued

Panel C: Subsample of firms controlled by non-diversified NTS holders

	(1)	(2)	(3)	(4)
Idiosyncratic risk	1.065*** [0.000]	1.055** [0.012]	0.751* [0.099]	0.768* [0.084]
Fraction of NTS	0.236*** [0.000]	0.239*** [0.000]	0.255*** [0.000]	0.255*** [0.000]
State NTS holding	0.045*** [0.000]	0.045*** [0.000]	0.044*** [0.000]	0.044*** [0.000]
ROA	-0.143*** [0.000]	-0.081 [0.002]	-0.082* [0.021]	-0.082* [0.021]
Batch	-0.002 [0.000]	-0.002*** [0.000]	-0.002*** [0.000]	-0.002*** [0.000]
Plan change	0.027*** [0.001]	0.027*** [0.001]	0.026*** [0.001]	0.026*** [0.001]
Additional promise	-0.012** [0.054]	-0.012** [0.054]	-0.009 [0.114]	-0.009 [0.114]
Sales growth	-0.011 [0.118]	-0.009 [0.182]	-0.010 [0.167]	-0.010 [0.167]
M/B	0.000 [0.958]	0.001 [0.725]	0.001 [0.779]	0.001 [0.779]
Stock return	-0.012 [0.300]	0.005 [0.704]	0.005 [0.710]	0.005 [0.710]
Leverage	0.012 [0.522]	0.019 [0.580]	0.019 [0.601]	0.019 [0.601]
Institutional TS holding	-0.105*** [0.000]	-0.105*** [0.000]	-0.104*** [0.000]	-0.104*** [0.000]
Foreign investor	0.042*** [0.000]	0.042*** [0.000]	0.042*** [0.000]	0.042*** [0.000]
CEO is COB	-0.009 [0.454]	-0.009 [0.454]	-0.009 [0.454]	-0.009 [0.454]
Board independence	0.049 [0.403]	0.049 [0.403]	0.049 [0.403]	0.049 [0.403]
Board size	0.000 [0.913]	0.000 [0.913]	0.000 [0.913]	0.000 [0.913]
Intercept	0.151*** [0.000]	0.143*** [0.000]	0.144*** [0.000]	0.127*** [0.000]
NTS holder, year, and industry FE	YES	YES	YES	YES
Number of observations	803	803	803	803
Adjusted R-squared	0.316	0.317	0.354	0.354

Panel B (diversified). In the diversified subsample, the coefficient on idiosyncratic risk is economically small and statistically insignificant. This matches the mechanism: diversified NTS holders already have risk-sharing through cross-firm portfolios, so tradability adds less marginal diversification value.

Panel C (non-diversified). In the non-diversified subsample, the idiosyncratic-risk coefficient is large and statistically significant (roughly 0.75–1.07 across columns). This is the sharpest evidence in favor of the risk-sharing interpretation (H2).

8.5 Table 4: testing the pure price-impact model (H3b) and falsification logic

Table 4 asks whether the *pure price impact* model (Eq. (3)) can reproduce compensation. The authors compute implied compensation ratios using two alternative proxies for PN :

- **Compensation Ratio B:** PN is net asset value per share (fiscal year end before reform).
- **Compensation Ratio C:** PN is set at an 80% discount to S (so $PN = 0.2S$), following prior work.

Table 4
Test of a pure price impact model

Panel A: Summary statistics

Variable	N	Mean	Std Dev	25 th Percentile	Median	75 th Percentile
Compensation ratio	1147	0.301	0.075	0.270	0.306	0.342
Compensation ratio B	1147	0.482	0.519	0.137	0.380	0.670
Compensation ratio C	1147	1.084	0.419	0.818	1.034	1.281

Panel B: Correlation matrix for different compensation ratios

	Compensation ratio	Compensation ratio B
Compensation ratio B	0.084*** [0.005]	1.000
Compensation ratio C	0.415*** [0.000]	0.523*** [0.000]

(continued)

Table 4
Continued

Panel C: Explaining the compensation ratio

	(1)	(2)	(3)	(4)
S/P_N	-0.002 [0.364]	-0.001 [0.611]	-0.001 [0.741]	-0.001 [0.690]
Fraction of NTS	0.258*** [0.000]	0.265*** [0.000]	0.281*** [0.000]	0.281*** [0.000]
Batch	-0.001*** [0.000]	-0.001*** [0.003]	-0.001*** [0.000]	-0.001*** [0.000]
Plan change	0.026*** [0.000]	0.026*** [0.000]	0.024*** [0.000]	0.025*** [0.000]
Additional promise	-0.017*** [0.000]	-0.014*** [0.002]	-0.011** [0.017]	-0.010** [0.018]
Sales Growth		-0.017*** [0.002]	-0.012** [0.023]	-0.013** [0.023]
Stock return		-0.025*** [0.006]	0.001 [0.894]	0.002 [0.880]
Leverage		0.037*** [0.007]	0.028** [0.036]	0.028** [0.039]
Institutional TS holding			-0.115*** [0.000]	-0.114*** [0.000]
Foreign investor			0.037*** [0.000]	0.037*** [0.000]
CEO is COB				0.006 [0.371]
Board independence				0.039 [0.427]
Board size				-0.000 [0.848]
Intercept	0.157*** [0.000]	0.135*** [0.000]	0.145*** [0.000]	0.132*** [0.000]
NTS holder, year, and industry FE	YES	YES	YES	YES
Number of observations	1147	1147	1147	1147
Adjusted R-squared	0.262	0.282	0.337	0.337

Panel A (levels). Both model-implied ratios are much larger than the actual mean compensation ratio, especially under the 80% discount calibration, indicating that the benchmark model does not match the level.

Panel B (correlations). The correlation between actual compensation and the model-implied ratio based on NAV is close to zero, while the discount-based implied ratio correlates more, but the overall mapping is still weak for a model meant to be mechanical.

Panel C (regression test, Eq. (5)). The regression

$$\text{CompensationRatio}_i = \alpha_0 + \beta_1(S/PN)_i + \beta_2\text{FractionNTS}_i + \dots + \varepsilon_i \quad (5)$$

yields a *negative and insignificant* coefficient on S/PN , contradicting the price-impact-only prediction (H3b). Hence, the authors argue that compensation cannot be explained solely by a simple price-impact compensation logic.

8.6 Table 5: robustness checks

Panel A (alternative samples). The main results are robust to:

- excluding “special treatment” (ST) firms,
- excluding firms with B-shares/H-shares (cross-listed or foreign-share classes),
- excluding the two pilot batches,
- restricting to cases where compensation is paid only in shares.

Table 5
Robustness check

Panel A: Using alternative samples

	(1)	(2)	(3)	(4)
Idiosyncratic risk	0.807** [0.019]	0.592* [0.074]	0.721* [0.071]	0.810** [0.014]
Fraction of NTS	0.268*** [0.000]	0.271*** [0.000]	0.245*** [0.000]	0.275*** [0.000]
State NTS holding	0.038*** [0.000]	0.040*** [0.000]	0.040*** [0.000]	0.044*** [0.000]
ROA	-0.151*** [0.000]	-0.092*** [0.028]	-0.093*** [0.030]	-0.118*** [0.001]
Batch	-0.002*** [0.000]	-0.001*** [0.000]	-0.001*** [0.000]	-0.001*** [0.000]
Plan change	0.024*** [0.000]	0.021*** [0.000]	0.020*** [0.000]	0.028*** [0.000]
Additional promise	-0.007* [0.093]	-0.006 [0.196]	-0.008 [0.079]	-0.007* [0.083]
Sales growth	-0.005 [0.360]	-0.009 [0.199]	-0.008 [0.175]	-0.004 [0.431]
M/B	-0.001 [0.861]	-0.000 [0.945]	-0.000 [0.919]	-0.000 [0.931]
Stock return	0.014 [0.170]	-0.002 [0.877]	-0.001 [0.961]	0.005 [0.593]
Leverage	-0.006 [0.625]	0.008 [0.560]	0.019 [0.219]	-0.005 [0.669]
Institutional TS holding	-0.097*** [0.000]	-0.084*** [0.000]	-0.100*** [0.000]	-0.091*** [0.000]
Foreign investor	0.033*** [0.000]	0.000 [0.966]	0.034*** [0.000]	0.036*** [0.000]
CEO is COB	0.006 [0.354]	0.006 [0.561]	0.004 [0.530]	0.004 [0.491]
Board independence	0.000* [0.054]	0.005 [0.380]	0.030 [0.536]	0.043 [0.366]
Board size	0.000 [0.757]	0.000 [0.747]	-0.000 [0.783]	0.001 [0.327]
Intercept	0.118*** [0.000]	0.120*** [0.000]	0.133*** [0.000]	0.101*** [0.000]
NTS holder, year, and industry FE	YES	YES	YES	YES
Number of observations	1048	1045	1110	991
Adjusted R-squared	0.405	0.348	0.357	0.410

Table 5
Continued

Panel B: Using alternative measures for NTS holders' bargaining power

	(1)	(2)
Idiosyncratic risk	0.823** [0.013]	0.757** [0.020]
Fraction of NTS	0.251*** [0.000]	0.274*** [0.000]
State NTS holding/Fraction of NTS	0.044*** [0.000]	
State is the largest NTS holder		0.027*** [0.000]
ROA	-0.084* [0.050]	-0.083* [0.051]
Batch	-0.002*** [0.000]	-0.001*** [0.000]
Plan change	0.022*** [0.000]	0.020*** [0.000]
Additional promise	-0.001 [0.801]	-0.009** [0.044]
Sales Growth	0.039 [0.133]	-0.009 [0.142]
MB	-0.000 [0.877]	-0.000 [0.907]
Stock return	0.004 [0.737]	0.003 [0.771]
Leverage	0.016 [0.288]	0.016 [0.286]
Institutional TS holding	-0.103*** [0.000]	-0.102*** [0.000]
Foreign investor	0.035*** [0.000]	0.035*** [0.000]
CEO is COB	0.007 [0.354]	0.006 [0.354]
Board independence	0.047 [0.362]	0.044 [0.362]
Board size	-0.000 [0.955]	-0.000 [0.853]

(continued)

Table 5
Continued

	(1)	(2)
Marketization	-0.001 [0.561]	-0.001 [0.584]
Intercept	0.134*** [0.000]	0.124*** [0.000]
NTS holder, year, and industry FE	YES	YES
Number of observations	1147	1147
Adjusted R-squared	0.359	0.365

Panel B (alternative bargaining-power measures and regional institutions). The paper replaces the bargaining proxy with:

- state NTS holding scaled by fraction NTS (to reduce collinearity with float expansion),
- an indicator that the state is the largest NTS holder.

It also includes a regional *marketization* index (institutional development). Main coefficients are stable; the marketization index is not statistically significant in these specifications.

9 Short summary

1. Compensation in China’s split-share privatization is not just about price impact; it also reflects risk-sharing gains for concentrated NTS holders.
2. Idiosyncratic risk (pre-reform) is a clean proxy for the marginal value of diversification for constrained owners; it predicts compensation in the cross-section.
3. The risk-sharing effect is stronger precisely where it should be strongest: when the controlling NTS holders are not diversified across firms.
4. A pure price-impact model has the right intuition for one dimension (float expansion), but fails on the key valuation-wedge implication (S/PN), suggesting missing economics in a price-impact-only view.
5. Empirically, the paper is a disciplined “institution + bargaining outcome” design: exploit a nationwide reform, measure equilibrium compensation, and map it to well-motivated primitives (risk, float, bargaining incentives, performance).

10 Conclusion

10.1 Summary

This paper studies China’s Split Share Structure Reform (2005–2007), a nationwide share-issue privatization episode in which previously non-tradable shares (held mainly by the state and legal persons) were converted into tradable shares. A central institutional feature is that conversion required a *negotiated compensation plan*: holders of non-tradable shares (NTS) compensated holders of tradable shares (TS) before their own shares could become tradable.

The paper’s core conclusion is that the observed compensation is not explained by a pure “price-impact” logic alone. Instead, compensation reflects **both**:

1. **Price impact** from a large increase in tradable float (more shares eventually coming to the market), and
2. **Risk-sharing gains** for NTS holders who were previously unable to trade and therefore could not diversify their concentrated equity positions.

Empirically, the compensation ratio is:

- **positively associated** with a proxy for *risk-sharing gains* (idiosyncratic risk),
- **positively associated** with a proxy for *price impact* (fraction of NTS / potential float expansion),
- **positively associated** with proxies for *weaker NTS bargaining power* (greater state NTS holdings, interpreted as a stronger political incentive to complete the reform),
- **negatively associated** with *firm performance* (e.g., ROA), consistent with better firms requiring less compensation to obtain approval.

A key reinforcing result is heterogeneity: the risk-sharing channel is **substantially stronger** in firms whose NTS holders are **not diversified** across many listed firms, and it is weak or absent where NTS holders are already diversified. This pattern supports the interpretation that the idiosyncratic-risk coefficient is capturing diversification motives rather than merely correlating with unobserved determinants of bargaining outcomes.

Finally, the paper concludes that risk sharing is an economically important efficiency gain in share-issue privatization: the reform did not merely “unlock float”; it also enabled a reallocation of idiosyncratic risk from concentrated controlling holders toward the broader market investor base.

10.2 Economic intuition: why compensation reveals risk-sharing gains

The intuition can be stated as one continuous chain:

1. **Pre-reform frictions created segmentation.** NTS holders owned large blocks but faced a binding *trading restriction*. This restriction prevented them from reducing exposure to firm-specific (idiosyncratic) risk.

2. **Tradability is a diversification technology.** Once NTS become tradable, concentrated owners can sell and diversify. The value of this new technology is larger when (i) the firm’s idiosyncratic risk is higher and/or (ii) the NTS holder is less diversified ex ante.
3. **Because TS holders have approval power, NTS holders must “pay” to obtain tradability.** The compensation ratio is the transfer that makes TS holders willing to accept the reform. Hence, compensation is an equilibrium object that shares the total surplus from removing the friction.
4. **Therefore, compensation increases with the value of tradability.** When risk-sharing gains are high (high idiosyncratic risk; non-diversified NTS holders), NTS holders are willing to offer more compensation to complete the reform. When float expansion is large (high fraction of NTS), TS holders demand more compensation to offset price-impact concerns.
5. **A pure price-impact benchmark misses an essential motive.** If compensation were only about offsetting dilution/price pressure, it should align tightly with the benchmark model’s key implication. The paper shows that the benchmark’s key valuation-wedge prediction is rejected in the data, while the risk-sharing patterns remain robust.

10.3 What to remember (compact takeaways)

- **Compensation is a revealed-preference object:** it embeds how market participants value removing a trading friction.
- **Risk sharing matters:** tradability allows concentrated owners to offload idiosyncratic risk to diversified market investors.
- **Heterogeneity is the smoking gun:** the risk-sharing effect appears where diversification is most valuable (non-diversified NTS holders).
- **Price impact still matters:** larger float expansions raise compensation, but cannot alone rationalize the compensation cross-section.

10.4 Takeaway

In China’s split-share privatization, negotiated compensation is jointly shaped by price-impact concerns and by risk-sharing gains from making concentrated, previously non-tradable holdings tradable, with the risk-sharing channel strongest when NTS holders are ex ante non-diversified.